

ELVIS YEBOAH-DUAKO

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Research Interest

Systems Security, Cloud and Serverless Security, Least-Privilege Access Control, AI Agent Security, Intent-Based Networking, Network Security, Information Flow Control, Data Privacy

Education

Ph.D. in Computer Science

2024 – Present

University of Massachusetts Amherst (Manning CICS), Amherst, MA

Advisor: [Professor Pubali Datta](#) | GPA: 3.74/4.0

M.Eng. in Computer Science and Technology

2021 – 2023

University of Electronic Science and Technology of China (UESTC), Chengdu, China

Thesis: *On Securing Resource-Constrained IoT Networks: A Provably Secure Blockchain-based Node Authentication and Key Agreement Protocol* | Advisor: [Professor Xia Qi](#) | GPA: 3.97/4.0

B.Sc. in Telecommunication Engineering

2016 – 2020

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

Advisor: [Dr. Emmanuel Ampoma Affum](#) | GPA: 3.86/4.0 | Best Graduating Student

Publications

Conference Papers

- [C1] D. Kang, **E. Yeboah-Duako**, S. Thomas and P. Datta. “When Apps Outlive Vendors: Security Implications of IoT Abandonware.” *ACM Conference on Computer and Communications Security (CCS)*, 2026. To appear. [\[Paper\]](#)

Under Review

- [S1] **E. Yeboah-Duako** and P. Datta. “Permissions on the Loose: Measuring Overprivilege in Real-World Serverless Applications.” *Under review, USENIX Security Symposium 2027*. [\[Paper\]](#)

Research Experience

Graduate Research Assistant (Ph.D.)

Fall 2024 – Present

Manning CICS, University of Massachusetts Amherst (Advisor: [Pubali Datta](#))

- **Least-privilege analysis policy for serverless applications [S1].** Designed and implemented a static-analysis framework that builds call graphs and performs data-flow analysis over AWS SDK calls to infer the IAM permissions each serverless function requires, compares them against granted policies, and recommends per-function least-privilege policies. Across 789 open-source applications, 46.7% required two or fewer permissions, yet 87.7% were granted three or more.
- **Validating LLM-generated network policies (PhD research milestone).** Building an AI-assisted Intent-Based Networking system (ONOS, OpenFlow, Mininet) in which LLMs compile high-level intents into executable flow rules. Characterizing failure modes of AI-generated policies and building an automated validation harness that checks deployed rules against intent-derived expectations for connectivity, reachability, and bandwidth via probe and live-network tests;
- **Authorization for AI agents (early-stage).** Investigating how authorization and access control

mechanisms can accommodate dynamic, tool-using agent behavior in cloud environments while preserving least privilege.

- **IoT abandonware [C1].** Co-led a study of the security risks that arise when IoT apps outlive their vendors (ACM CCS 2026). Led the blocklist analysis of embedded resources and the domain-ownership change analysis parts, and contributed to the drafting of the paper and its related-work survey.

Graduate Research Assistant (M.Eng)

2021 – 2023

Institute for Blockchain & Security, UESTC, China

- **M.Eng. thesis.** Designed a lightweight ECC-based authentication and key-agreement protocol for resource-constrained IoT networks, with blockchain-based identity management. Proved session-key security via a game-based reduction in the Real-or-Random model (ECDDH assumption), complemented by informal analysis against replay, man-in-the-middle, impersonation, and ephemeral-key-leakage attacks; in simulation, constrained nodes completed mutual authentication in under 100 ms.
- Conducted systematic literature reviews on secure blockchain applications in human-resources management and contributed to manuscript preparations for senior colleagues.

Research Assistant

2020 – 2021

Department of Telecommunication Engineering, KNUST, Ghana

- Evaluated deep-learning and classical machine-learning approaches for DDoS detection in SDN/NFV environments, designing network simulations to assess architecture resilience under attack.

Industry Experience

Software Engineer

2022 – 2024

Orgmento Ltd., Ghana (Subcontractor: Cambridge Crystallographic Data Centre (CCDC), Cambridge, UK)

- Led a phased identity-modernization project where I migrated 60,000+ user profiles from a legacy in-house Identity provider to Azure AD B2C without password resets. Subsequently built an identity gateway providing OpenID Connect and SAML-based SSO across CCDC's tools and services.
- Evaluated and integrated GitHub Advanced Security (CodeQL static analysis) and SonarCloud into CI pipelines to detect vulnerabilities and enforce code-quality gates for software shipped by CCDC and Orgmento.
- Developed a web platform that lets research scientists and non-technical users to run CCDC-engineered research scripts and interactively explore experiment results without managing the underlying code.

Technical Skills

Languages: Python, C#, Go, JavaScript/TypeScript, C, Datalog

Program Analysis: call-graph and data-flow analysis, IAM policy analysis, CodeQL

Cloud & Identity: AWS (IAM, Lambda), Azure (AD B2C), Serverless Framework, OpenID Connect

Networking: SDN, OpenFlow, ONOS, Mininet

Talks

Enforcing Least-Privilege Authorization in Serverless Cloud Applications

Spring 2025

Guest Speaker, Women in Cybersecurity CTF, UMass Amherst Chapter

Awards & Honors

Inclusive Leadership Fellowship, UMass Amherst

Fall 2025

Jim Gray Scholarship in Computer Science, UMass Amherst	<i>Spring 2025</i>
Spaulding-Smith Fellowship, UMass Amherst	<i>Fall 2024</i>
UESTC University Graduate Scholarship Award	<i>Fall 2021</i>
Best Instructor, Huawei ICT Academy Ambassador's Award	<i>Spring 2021</i>
Best Graduating Student, Telecommunication Engineering, KNUST	<i>Fall 2020</i>
Outstanding Prize, Huawei ICT Competition — Sub-Saharan Africa Regional Finals	<i>Fall 2020</i>
First Prize, Huawei ICT Competition — National Finals, Ghana (Network Track)	<i>Fall 2019</i>

Teaching Experience

Teaching Assistant, COMPSCI 453: Computer Networks, UMass Amherst	<i>Fall 2026</i>
Teaching Assistant, COMPSCI 560: Intro. to Computer & Network Security, UMass Amherst	<i>Fall 2025</i>
Teaching Assistant, Mobile Computing, UESTC, China	<i>Spring 2022</i>
Teaching Assistant, TE 453: Computer Networking, KNUST, Ghana	<i>Fall 2021</i>
Instructor, Huawei ICT Academy, KNUST, Ghana	<i>Spring 2021</i>

Service & Mentoring

Graduate Employee Organization (GEO) Steward for Manning CICS, UMass Amherst	<i>Fall 2026</i>
New Student Committee, Manning CICS, UMass Amherst	<i>2026/2027</i>
Judge, Massachusetts Science & Engineering Fair, Middle School Division	<i>Spring 2026</i>
Graduate Mentor, IDS STEMPOWER Fellowship Program, UMass Amherst	<i>2025/2026</i>
PhD Peer Mentor, Manning CICS, UMass Amherst	<i>2025/2026</i>